Title: SUNLIGHT TRACKING SYSTEM USING ARDUINO UNO

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Empathize

Problem identification

❖Solar panels fixed at one position do not get maximum sunlight all the time.

❖Sun moves from east to west → fixed panel produces less power.

❖Energy efficiency reduces due to poor alignment.

User needs

❖A system that automatically adjusts towards sunlight.

❖Low-cost and easy-to-build solar tracking method.

❖Improved solar power output without human effort.

❖Works reliably throughout the entire day.

Define

Problem statement

❖To design a system that detects sunlight direction using sensors.

❖To rotate the solar panel using a servo or DC motor.

❖To ensure the panel continuously faces the brightest direction.

❖To increase solar energy absorption efficiently.

Objectives

❖Sense sunlight using LDR sensors.

❖Compare light intensity on left and right sides.

❖Track sunlight automatically from morning to evening

Ideate

Possible solutions

❖Use two LDR sensors to detect sunlight difference and rotate the panel.

❖Use four LDR sensors for more accurate sunlight detection.

❖Use a single-axis servo motor to follow the sun east to west.

❖Use a dual-axis system to track sun movement in all directions

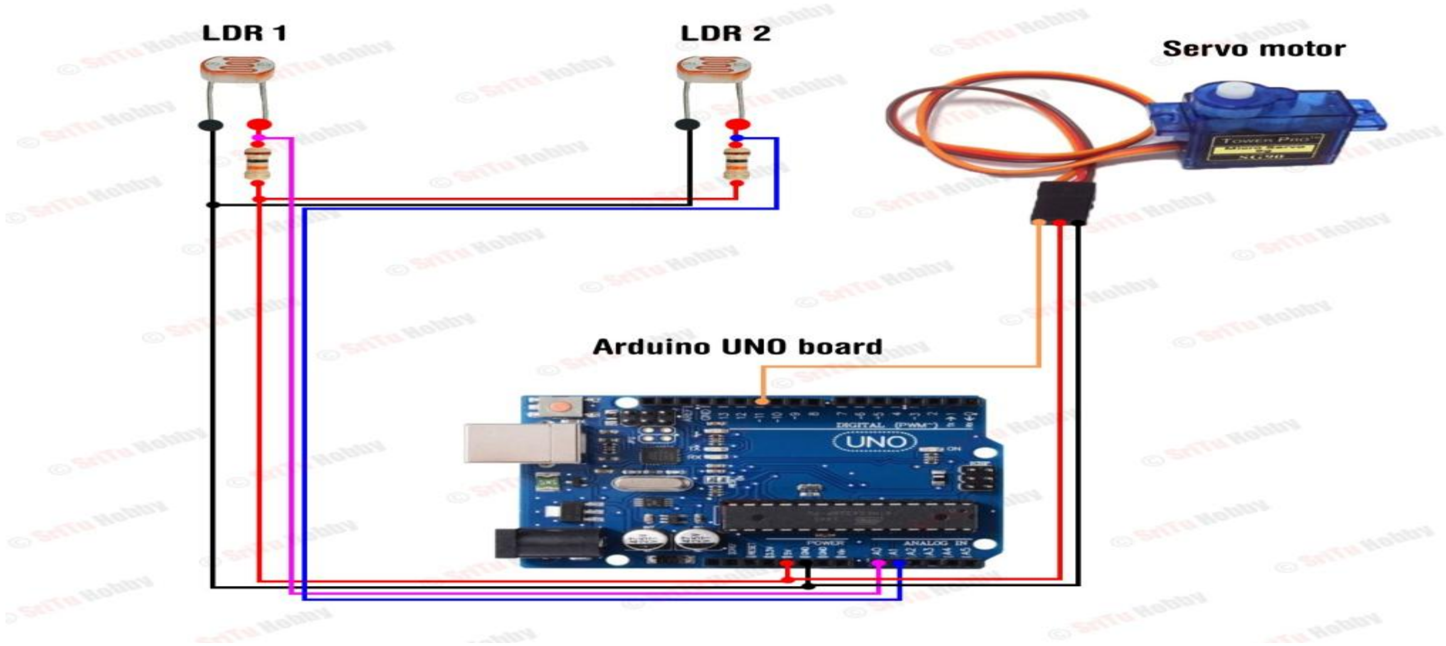
Concept selection

❖**Arduino UNO** → easy to program.

❖**Servo Motor** → precise movement.

❖**Low cost** and suitable for students.

Prototype



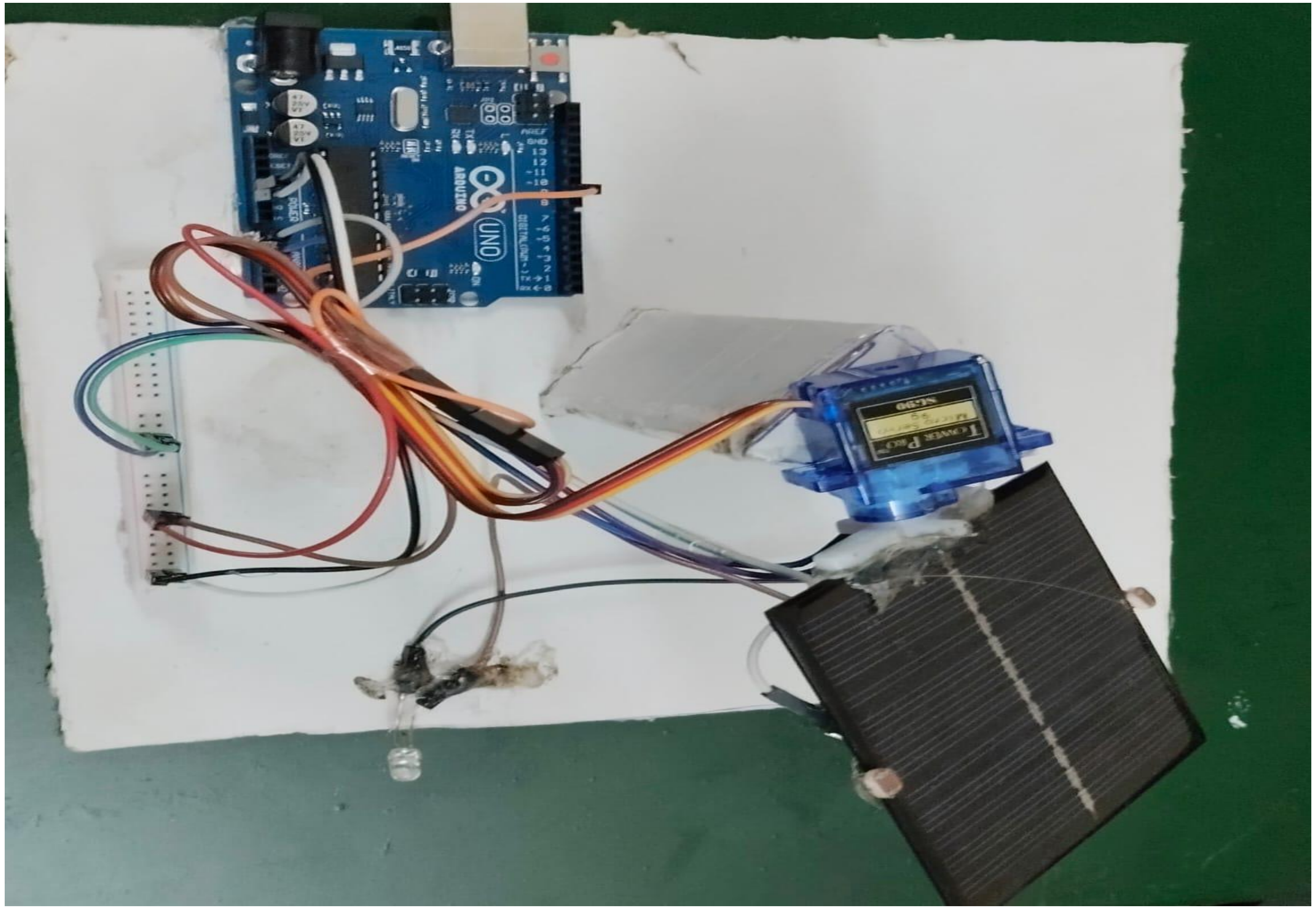
Hardware/software details

❖Arduino UNO is used as the main controller for reading sensors and controlling the motor.

❖Two LDR sensors are connected to the analog pins to sense sunlight intensity.

❖10kΩ resistors are used with LDRs to form voltage divider circuits

Test



Conclusion & Future scope

- ❖The system successfully tracks sunlight using LDR sensors and Arduino.
- ❖It increases the efficiency of solar panels by maintaining the best angle.
- ❖The design is simple, low-cost, and easy to implement.
- ❖Can be upgraded to dual-axis tracking for higher efficiency.
- ❖Can integrate IoT to monitor solar panel performance remotely.
- ❖Can use ML/AI to predict sun position even during cloudy weather

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Thank you